

CARLOS C. TAPANG

SENIOR SOFTWARE ENGINEER / LEADER

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Software engineer with 20+ years of experience in establishing technical direction, driving technology/data strategy and solutions architecture, leading global teams, and managing complex project lifecycles to develop, deliver, and debug cutting-edge SaaS and cloud applications.

- Proven track record of leading and aligning people, processes, and technologies across the software development life cycle (SDLC) to drive on-time, on-budget delivery of scalable, high-quality applications and systems that deliver exceptional user experience.
- Extensive background in fullstack software development for both web and mobile applications in Agile environments using either Microsoft or Linux/Node/MySQL/React (LiNoMyRe) stacks.
- Excels in managing and mentoring collaborative, cross-functional teams and advising leaders across the business and technology functions to accelerate product velocity, foster technology innovation, and improve service quality.

CORE COMPETENCIES

Full-Stack Software Development | Software Engineering | Solutions Architecture | Linux / Node / MySQL / React | React Native | C#, .NET, JavaScript, TypeScript | Coding & Code Review | Web & Mobile Application Development | Systems Integration | SDLC Management | Cloud Technologies (AWS, Azure) | Database Design & Administration | Continuous Integration & Continuous Delivery (CI/CD) | Agile Scrum Methodologies | Test Automation | Process Improvement | Team Management | Training & Mentoring | Cross-Functional Collaboration

PROFESSIONAL EXPERIENCE

Senior Software Engineer | City of Auburn, Auburn, WA

Apr 2023 to Present

- Troubleshooted and fixed issues with Praxis (a web-based city tax calculator and tracker), in addition to developing and implementing new features requested by end users.

Application Development Lead (Part-Time) | RockStable Token, Bellevue, WA

Jan 2021 to Present

- Led a 4-member development team in developing, managing, and maintaining a remittance application that served foreign nationals (primarily in the Philippines) who had traveled to the US on work visas.
- Coded the frontend in React and Node.js and the backend in Docker and Kubernetes, including running a web server Express on top of Node.js and creating backend APIs for React with Node.js.

Lead Fullstack Developer | Microsoft, Redmond, WA

Mar 2020 to Dec 2020

- Developed various ecommerce applications for the Microsoft Universal Store using Angular 5 for the frontend and .NET, SQL, and C# for the backend, in addition to serving on an Agile team that maintained the Huge Partnership Connection web app.

Senior Frontend React Developer | Qualtrics, Seattle, WA

Oct 2019 to Dec 2019

- Architected web-based analytical and data tracking applications (including building the front-end customer-facing UI from scratch) while working on an Agile team to develop the Qualtrics Participant Info web app.

Senior Software Developer | Senior Full-Stack Developer | Microsoft, Redmond, WA

Apr 2017 to Jun 2019

- Led the development of automated reports in Dynamics CRM that compared Dynamics data with actual App Insight telemetry data, involving the creation of complex programs in Logic Apps and the use of Azure to build the reports.
- Used React and React.js (including React-Redux 6.0.3) to code applications on the frontend, in addition to building REST and Web APIs and fixing client-side bugs using BotDesigner.
- Leveraged Power BI to use App Insight data for analysis and utilized Graph API to successfully graph data on a large scale down to very specific objects, enabling successful development of error reports.
- Drove maintenance and feature enhancement efforts for the Microsoft Security Response Center portal's web UI using Angular.js, fixing all accessibility issues, automating the processing of localized resources, and removing jQuery from the code.

Software Design Engineer 3 | Nordstrom Tech, Seattle, WA

Apr 2016 to Mar 2017

- Engineered an AWS application for direct marketing analytics, using AWS Lambda functions, Redshift, and REST APIs to extract reports from 5 marketing vendors and enable analysis of campaign effectiveness by linking clicks and impressions to sales.
- Served as the key design engineer for Nordstrom's Inventory Deployment Enterprise App, improving app reliability and maintainability through code refactoring, class structure optimization, and implementation of caching mechanisms using MemoryCache.

Lead Developer | Senior Software Design Engineer | Microsoft, Redmond, WA

Oct 2014 to Apr 2016

- Led an offshore team of 10 engineers in implementing devicesupport.microsoft.com, a vital device registration and support platform for Surface, Xbox, and Microsoft Band users handling 5K+ daily visitors.

- Authored a comprehensive Top Level Design Document (TLD) for MS Turn 10 Studios to seamlessly integrate the camera control tool within an enhanced workflow framework, optimizing efficiency and usability.
- Designed WPF applications utilizing the MVVM pattern to enable game artists to capture snapshots and videos of the Forza Motorsport game, including writing code for SCRUM backlog item implementation, bug resolution, and overall game shipping improvements.

Senior Backend Developer | Centerus, Inc., Bellevue, WA

Mar 2014 to Aug 2014

- Led the architecture and development of a Bitcoin mining card in just 6 months, building the firmware for its PIC32 microcontroller using Razor, C#, .Net, and MVC and implementing a simple, interrupt-driven system without an OS to build functional components.

Backend Developer | Synergy Sports Tech, Bellevue, WA

Oct 2013 to Mar 2014

- Architected and developed the Synergy 360 Wiki Product, enhancing scalability by implementing ASP.Net async constructs and enabling a system originally designed for 100 users to efficiently accommodate up to 1K concurrent users.

Senior Frontend Developer | Microsoft, Redmond, WA

Jul 2013 to Sep 2013

- Collaborated within an Agile team to develop media and entertainment tools for Xbox and 2-way TV producers, employing the MVVM design pattern in WPF and making intensive use of code reviews to accelerate development of highly reliable code.

Senior Full-Stack Developer | Intel Corporation, Chandler, AZ

Nov 2012 to Jan 2013

- Fixed bugs in a WPF application designed in the MVVM pattern, performing heavy frontend coding in WPF and backend coding in .NET and C# in an Agile Scrum environment using Visual Studio TFS and Rally.

Senior Backend Developer | Orion's Digital Systems (ODS), Bellevue, WA

May 2011 to Oct 2012

- Resolved complex compatibility issues in ODS media handlers and integrated the ODS asset system with media handling interfaces, involving a rebuild of MP4 and H.264 media handlers using Design Patterns to ensure ease of use.
- Designed 4 media handling interfaces (MP4, Microsoft ISMV, Apple movie, and Android MP4 media) in C# with Linq and anonymous type extensions, in addition to building several media handlers that implemented the same interfaces.
- Contributed to the development of multiple Silverlight applications that used the media handling interfaces as well as Mpeg-4, H.264, and MJPEG decoders for web random access and video fragmentation libraries for Silverlight and WPF applications.

Frontend Mobile Architect | Microsoft, Redmond, WA

Dec 2010 to Mar 2011

- Created a prototype Silverlight app for the iPhone in less than 4 months, completing just the necessary classes in Silverlight to run a simple Windows Mobile app on iOS.

President & Founder | Centerus, Inc., Seattle, WA

Jan 2009 to Nov 2010

- Developed and released 5 news/entertainment podcast apps on the iPhone using Apple's Objective-C, including publishing a game app for the iPhone (SlasherDerby) using JavaScript on a Mono game engine.

ADDITIONAL EXPERIENCE

Software Sustaining Engineer | Software Design Engineer II | Senior Software Engineer | Microsoft, Redmond, WA

- Developed a middle-layer module for a project with Lego utilizing C#, an MS Records system using Office Server 2007 in collaboration with Wipro, a WebDAV IHttpHandler using Design Patterns in C#, and a massive SQL Server database with ASP.Net UI.
- Resolved 5 bugs in the Windows XP Tablet PC Edition, in addition to developing a Windows command-line script (CLS) program for hotfix builds (CMD script).
- Retrofitted modules for logging functionality to Visual Studio .NET (Everett), including a C# parser for log file scanning and SQL Server database entry, a C# utility for log file extraction to a local server, and a JavaScript module for periodic execution.

Software Engineer | Intel, Hillsboro, OR

- Led a 7-person team in developing a COM/DCOM-based system management tool using Visual C++/Windows, implementing a semi-automated capability to report application problems across multiple Intel divisions to a service site.
- Managed the Internet multimedia Trials project for Intel Architecture Labs, installing routers and internet links to connect 4 Intel sites through an intranet to transmit and route multimedia streams across 4 states.
- Guided a 4-person team in the Network Products Division, focusing on developer testing and bug fixes for support and maintenance of NIC drivers for Windows 95 and Windows NT.
- Conducted testing for the iRMX 86 Real-Time Operating System, including building device drivers and a Unix file interface for iRMX as well as a debugger interface to a C compiler for 80386.

EDUCATION

Bachelor of Science (BS) in Physics | Ateneo de Manila University, Quezon City, Philippines

Certificate of Completion in Microcomputer Research | Tokyo University, Tokyo, Japan